

Deploying and Securing Kubernetes with NGINX, Prometheus & Grafana on University Infrastructure

Prerequisites

- Kubernetes cluster (Kind or any other Kubernetes cluster)
- Helm package manager
- Basic knowledge of Kubernetes and Prometheus

Project setup :

We will be using a virtual machine in the faculty's cloud. Customise the openssh config file

```
└──(kali㉿kali)-[~/ssh]
```

```
└─$ cat config
```

Configuration for the fep.grid.pub.ro jump host

Host fep

HostName fep.grid.pub.ro

User mohamed.tlili

IdentityFile ~/.ssh/id_openstack2

ServerAliveInterval 100

ForwardX11 yes

Configuration for the OpenStack VM, accessed via the fep jump host

Host openstack-vm

HostName 10.9.4.118

User student

IdentityFile ~/.ssh/id_openstack2

ProxyJump fep

ServerAliveInterval 100

ForwardAgent yes

Host kubernetes-vm

HostName 192.168.100.51 # Accessible IP of the Kubernetes VM

User student

ProxyCommand ssh -W %h:%p openstack-vm

LocalForward 9090 localhost:9090

LocalForward 3000 localhost:3000

LocalForward 30088 localhost:30088

LocalForward 9113 localhost:9113

LocalForward 9114 localhost:9114

LocalForward 30080 localhost:30080 Then :

Ssh kuberntes-vm

In the Base Virtual Machine :

Ssh to open stack vm **Download the project archive cd ~/work wget**

<https://repository.grid.pub.ro/cs/scgc/laboratoare/homework-project.zip>

unzip homework-project.zip **Start the virtual machines bash runvm.sh virsh net-dhcp-leases labvms**

****Resize disk and filesystem****

Ssh 192.168.100.51 sudo

growpart /dev/sda 2

sudo resize2fs

/dev/sda2 df -h | grep

/dev/sda2 **Manifestes**

****used : nginx-****

deployment.yaml :

apiVersion: apps/v1

kind: Deployment

metadata: name: nginx

```

namespace: monitoring
spec:
  replicas: 1 selector:
matchLabels:      app:
nginx template:
metadata:         labels:
app: nginx        spec:
containers:       - name:
nginx             image:
nginx:latest      ports:
                  - containerPort: 80
containerPort: 8080 volumeMounts:
- name: html        mountPath:
/usr/share/nginx/html - name:
nginx-config        mountPath:
/etc/nginx/nginx.conf subPath:
nginx.conf          volumes:      - name:
html               configMap:
                  name: nginx-html
- name: nginx-config
configMap:
  name: nginx-configmap
  nginx-
configmap.yaml :
apiVersion: v1 kind:
ConfigMap metadata:
  name: nginx-configmap
namespace: monitoring
data:

```

```

    nginx.conf: |      events {}
http {      server {      listen
80;      location / {
root /usr/share/nginx/html;
      }      }
server {      listen
8080;      location
/metrics {
stub_status;
      }
      }
    }
    nginx-html-

```

configmap.yaml :

apiVersion: v1 kind:

ConfigMap metadata:

name: nginx-html

namespace: monitoring

data:

index.html: |

```
<html><body><h1>Welcome to NGINX!</h1></body></html>
```

index.html

:

```
<html><body><h1>Welcome to NGINX!</h1></body></html>
```

nginx-prometheus-exporter-

deployment.yaml :

```

apiVersion: apps/v1 kind:
Deployment metadata:
  name: nginx-prometheus-exporter
namespace: monitoring spec:
  replicas: 1
selector:
matchLabels:
  app: nginx-prometheus-exporter
template:
metadata:
labels:
  app: nginx-prometheus-exporter
spec:
  containers:
- name: nginx-prometheus-exporter      image:
  nginx/nginx-prometheus-exporter:latest      args:
- --nginx.scrape-uri=http://nginx:8080/metrics
- --web.listen-address=:9114      ports:
- containerPort: 9114
  nginx-prometheus-exporter-service.yaml
:
apiVersion: v1 kind:
Service metadata:
  name: promexporter
namespace: monitoring
spec:
  selector:
    app: nginx-prometheus-exporter
ports:
  - protocol: TCP
port: 9114      targetPort: 9114

```

```

    nginx-prometheus-exporter.yaml
: apiVersion: apps/v1 kind:
Deployment metadata:
    name: nginx-prometheus-exporter
namespace: default spec:
    replicas: 1
selector:
matchLabels:
    app: nginx-prometheus-exporter
template:      metadata:
labels:
    app: nginx-prometheus-exporter
spec:      containers:
- name: nginx-prometheus-exporter      image:
    nginx/nginx-prometheus-exporter:latest      args:
- --nginx.scrape-uri=http://nginx:8080/metrics
- --web.listen-address=:9114      ports:
- containerPort: 9114
    nginx-service.yaml
:
apiVersion: v1
kind: Service
metadata:
name: nginx
namespace:
default spec:
selector:
app: nginx
ports:      -

```

```
name: http
protocol: TCP
port: 80
targetPort: 80
nodePort: 30080
- name: metrics
protocol: TCP
port: 8080
targetPort:
8080
nodePort: 30088
type: NodePort
```

Customised part prometheus config map :

```
prometheus.yml: |
global:
    scrape_interval: 15s
scrape_configs:
-
job_name: prometheus
static_configs:
- targets:
- localhost:9090      - job_name: nginx      static_configs:
- targets:
- promexporter.monitoring.svc.cluster.local:9114
```

grafana-config.yaml :

apiVersion: v1 kind:

ConfigMap metadata:

```
name: grafana-config
namespace: monitoring
data: grafana.ini: |
    [security]  admin_user =
admin  admin_password =
AiRukqX0Su
```

Steps Taken:

Step 1: Create a Kubernetes Cluster kind

```
create cluster
```

Step 2: Deploy NGINX Service

1. Create and apply the NGINX ConfigMap:

```
nano nginx-configmap.yaml kubectl
apply -f nginx-configmap.yaml
```
2. Create and apply the NGINX HTML ConfigMap:

```
nano nginx-html-configmap.yaml
kubectl apply -f nginx-html-configmap.yaml
```
3. Create and apply the NGINX Deployment:

```
nano nginx-deployment.yaml
kubectl apply -f nginx-deployment.yaml
```

4. Create and apply the NGINX Service:

```
nano nginx-service.yaml
kubectl apply -f nginx-service.yaml
```

Step 3: Deploy NGINX Prometheus Exporter

5. Create and apply the NGINX Prometheus Exporter Deployment:

```
nano nginx-prometheus-exporter.yaml
kubectl apply -f nginx-prometheus-exporter.yaml
```


6. Create and apply the NGINX Prometheus Exporter Service:

```
nano nginx-prometheus-exporter-service.yaml
kubectl apply -f nginx-prometheus-exporter-service.yaml
```

Step 4: Deploy Prometheus

7. Add Prometheus Helm repository:

```
helm repo add prometheus-community
https://prometheuscommunity.github.io/helm-charts
```

8. Install Prometheus using Helm: `helm install prometheus prometheus-community/prometheus --namespace monitoring`

9. Create and apply the Prometheus ConfigMap:

```
nano prometheus-config.yaml
kubectl apply -f prometheus-config.yaml
```

Step 5: Deploy Grafana

10. Add Bitnami Helm repository:

```
helm repo add bitnami https://charts.bitnami.com/bitnami
```

11. Install Grafana using Helm:

```
helm install grafana bitnami/grafana --namespace monitoring
```

12. Create and apply the Grafana ConfigMap:

```
nano grafana-config.yaml
kubectl apply -f grafana-config.yaml
```

Step 6: Port Forwarding

13. Forward ports to access services:

```
kubectl port-forward svc/nginx 30080:80 -n monitoring kubectl
port-forward svc/nginx 9114:80 -n monitoring kubectl port-
forward svc/prometheus-server 9090:80 -n monitoring kubectl
port-forward svc/grafana 3000:80 -n monitoring
```

Step 7: Perform DoS Attack

14. Run the DoS attack using **slowhttptest**: `slowhttptest -c 1000 -H -i 10 -r 200 -t GET -u http://127.0.0.1:30080 -x 24 -p 3` **Observations:**

- Before the attack, the server handled connections smoothly with low active connections and consistent request rates.
- During the attack, there was a noticeable increase in active and handled connections, reflecting the server's struggle to keep up with the load.
- After the attack, the server failed to respond, and the metrics dropped, highlighting the server's inability to recover from the overload

NGINX Metrics :

- **System Behavior:**
 - **Response Time:** Increased due to the high load, leading to slower responses or timeouts.
 - **Service Availability:** The NGINX server became unresponsive after a certain point, resulting in connection refusals.

Impact:

The DoS attack significantly impacted the NGINX server's ability to handle connections. This was evident from the spike in the metrics during the attack and the subsequent drop to zero, indicating that the server was unable to process requests.

The services (NGINX, Prometheus, and Grafana) crashed or became unresponsive under the high load, as observed by zero values in the metrics post-attack and the need for service restarts.

Conclusion:

- Under high load, the NGINX server struggled to maintain performance and eventually became unresponsive, leading to service crashes.
- Preventive measures such as rate limiting, resource limits, and scaling replicas are essential to mitigate the impact of high load scenarios

Step 8: Verify Setup

15. Access NGINX, Prometheus, and Grafana in the browser:

- NGINX: <http://localhost:30080> ○ NGINX

metrics: <http://localhost:9114> ○ NGINX

metrics: <http://localhost:30088> ○

Prometheus: <http://localhost:9090> ○ Grafana:
<http://localhost:3000>

Task 6 - Security-Related Actions

Proposed Security-Related Actions to Secure the Web Server and Infrastructure:

Rate Limiting: ○ Implement rate limiting on the NGINX server to prevent abuse and mitigate DoS attacks. ○ Example configuration in `nginx.conf`:

```
http
{
    limit_req_zone $binary_remote_addr zone=mylimit:10m rate=1r/s;
    server {
        location / {
            limit_req zone=mylimit
            burst=10;
            proxy_pass http://backend;
        }
    }
}
```

Resource Limits and Requests:

○ Define resource limits and requests for the NGINX, Prometheus, and Grafana deployments to ensure fair usage of resources and prevent a single service from exhausting cluster resources. ○ Example configuration in deployment YAML:

```
resources:
limits:
    cpu: "500m"
memory: "512Mi"
requests:    cpu:
"250m"      memory:
"256Mi"
```

Replicas and Load Balancing: ○ Increase the number of replicas for each deployment and use a load balancer to distribute traffic evenly across replicas. ○ Example command to scale deployment:

```
kubectl scale deployment nginx --replicas=3 -n monitoring
```

Network Policies: ○ Implement network policies to control the flow of traffic between pods and services, restricting access to only necessary components. ○ Example network policy:

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy metadata:
name: allow-nginx namespace:
monitoring spec: podSelector:
matchLabels: app: nginx
policyTypes: - Ingress -
Egress ingress:
  from:
  podSelector:
  matchLabels:
    app: prometheus
egress:
  to:
  podSelector:
  matchLabels:
    app: prometheus
```

TLS and Encryption:

- Use TLS to encrypt traffic between clients and the NGINX server to protect data in transit.
- Obtain and configure a TLS certificate for the NGINX server.

Monitoring and Alerts:

- Configure Prometheus to monitor critical metrics and set up alerts to notify administrators of potential issues such as high load or resource exhaustion. ○ Example alert rule in Prometheus: **groups:** - **name:**

```
example rules:
alert: HighCPUUsage
expr:
sum(rate(container_cpu_usage_seconds_total{image!=""}[1m])) by
(pod_name) > 0.8
for: 2m
labels:
  severity: warning
annotations:
```

summary: "High CPU usage detected"
usage is above 80% for more than 2 minutes."
description: "CPU

By implementing these security measures, the web server and the entire infrastructure will be more resilient to attacks and capable of maintaining performance and availability under various load conditions.

Screenshots :

```

(kali@kali)-[~]
└─$ ssh kubernetes-vm

* IMPORTANT - PLEASE READ CAREFULLY:
* Începând cu 28.09.2023 vom muta directoarele home vechi la calea $OLD_HOME. Vă rugăm să mutați
ce mai este relevant în home-ul cel nou pana pe 05.10.2023
* If you are using GPU computing and TensorFlow technology, please use CUDA_VISIBLE_DEVICES
to limit the number of GPUs. Otherwise, you will be banned.
* Your sessions will be disconnected automatically after 60min on inactivity.
You can use "-o ServerAliveInterval 10" option if you need longer sessions.*
(mohamed.tlili@fep.grid.pub.ro)

Please login at https://login.upb.ro/auth/realms/UPB/device. Then input code OUGM-YPLI

Press Enter to continue:
Enter passphrase for key '/home/kali/.ssh/id_openstack2':
student@192.168.100.51's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-174-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Fri 24 May 2024 04:46:36 PM UTC

System load:                0.04
Usage of /:                  34.1% of 15.64GB
Memory usage:               59%
Swap usage:                  0%
Processes:                   184
Users logged in:             0
IPv4 address for br-ca31d26d4b3e: 172.18.0.1
IPv6 address for br-ca31d26d4b3e: fc00:f853:ccd:e793::1

student@hw-mohamed-abdelhamid-tlili:~/work$ wget https://repository.grid.pub.ro/cs/scgc/laboratoare/homework-project.zip
--2024-05-08 13:36:10-- https://repository.grid.pub.ro/cs/scgc/laboratoare/homework-project.zip
Resolving repository.grid.pub.ro (repository.grid.pub.ro) ... 141.85.241.222
Connecting to repository.grid.pub.ro (repository.grid.pub.ro)|141.85.241.222|:443 ... connected.
HTTP request sent, awaiting response ... 200 OK
Length: 147141980 (140M) [application/zip]
Saving to: 'homework-project.zip'

homework-project.zip 100%[=====>] 140.33M 141MB/s in 1.0s

2024-05-08 13:36:11 (141 MB/s) - 'homework-project.zip' saved [147141980/147141980]

student@hw-mohamed-abdelhamid-tlili:~/work$ unzip homework-project.zip
Archive: homework-project.zip
  creating: lab-kubernetes/
  inflating: lab-kubernetes/checksums
  inflating: lab-kubernetes/lab-kubernetes.qcow2
  inflating: lab-kubernetes/config
  inflating: runvm.sh
student@hw-mohamed-abdelhamid-tlili:~/work$ bash runvm.sh
Network labvms defined from /usr/share/libvirt/networks/labvms.xml

Network labvms marked as autostarted

Network labvms started in openstack2 repo

net.ipv4.ip_forward = 1
Updated network labvms live state
Image resized.

Starting install...
Domain creation completed.
student@hw-mohamed-abdelhamid-tlili:~/work$ virsh net-dhcp-leases labvms
Expiry Time    MAC address    Protocol    IP address    Hostname    Client ID or DUID

```

```
student@hw-mohamed-abdelhamid-tlili:~/work$ ssh 192.168.100.51
The authenticity of host '192.168.100.51 (192.168.100.51)' can't be established.
ECDSA key fingerprint is SHA256:sg451Iub10EY2jnRt7qt1mpbx4Cc+Wa2NrOne/wZUts.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.100.51' (ECDSA) to the list of known hosts.
student@192.168.100.51's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-174-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

System information as of Wed 08 May 2024 01:37:26 PM UTC

System load:	1.72	Processes:	124
Usage of /:	16.8% of 7.77GB	Users logged in:	0
Memory usage:	10%	IPv4 address for docker0:	172.17.0.1
Swap usage:	0%	IPv4 address for eth0:	192.168.100.51

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.
To check for new updates run: `sudo apt update`

```
student@lab-kubernetes:~$
```

```
student@lab-kubernetes:~$ sudo growpart /dev/sda 2
CHANGED: partition=2 start=4096 old: size=16771072 end=16775168 new: size=33550303 end=33554399
student@lab-kubernetes:~$ sudo resize2fs /dev/sda2
resize2fs 1.45.5 (07-Jan-2020)
Filesystem at /dev/sda2 is mounted on /; on-line resizing required
old_desc_blocks = 1, new_desc_blocks = 2
The filesystem on /dev/sda2 is now 4193787 (4k) blocks long.
```

```
student@lab-kubernetes:~$ df -h | grep /dev/sda2
/dev/sda2    16G  1.4G   14G   9% /
student@lab-kubernetes:~$
```

```
student@lab-kubernetes:~$ kind create cluster --name my-cluster
```

Creating cluster "my-cluster" ...

✓ Ensuring node image (kindest/node:v1.29.2) 🖼️

✓ Preparing nodes 📦

✓ Writing configuration 📄

✓ Starting control-plane 🏹 `uncrack2rep`

✓ Installing CNI 🌐

✓ Installing StorageClass 💾

Set kubectl context to "kind-my-cluster"

You can now use your cluster with:

```
kubectl cluster-info --context kind-my-cluster
```

Have a question, bug, or feature request? Let us know! <https://kind.sigs.k8s.io/#community> 🙋

```
student@lab-kubernetes:~$
```

```
student@lab-kubernetes:~$ nano nginx-html.yaml
```

```
student@lab-kubernetes:~$ kubectl apply -f nginx-html.yaml
```

configmap/nginx-html created

```
student@lab-kubernetes:~$ nano nginx-deployment.yaml
```

```
student@lab-kubernetes:~$ kubectl apply -f nginx-deployment.yaml
```

The Deployment "nginx" is invalid:

* spec.template.spec.volumes[1].configMap.items[0].path: Required value

* spec.template.spec.containers[0].volumeMounts[1].name: Not found: "nginx-conf-vol"

```
student@lab-kubernetes:~$ nano nginx-deployment.yaml
```

```
student@lab-kubernetes:~$ kubectl apply -f nginx-deployment.yaml
```

deployment.apps/nginx created

```
student@lab-kubernetes:~$ nano nginx-service.yaml
```

```
student@lab-kubernetes:~$ kubectl apply -f nginx-service.yaml
```

service/nginx created


```
student@lab-kubernetes:~$ sudo systemctl start nginx
student@lab-kubernetes:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable nginx
student@lab-kubernetes:~$ sudo yum update -y
sudo: yum: command not found
student@lab-kubernetes:~$ sudo nano /etc/nginx/conf.d/your_site.conf
student@lab-kubernetes:~$ sudo nano /etc/nginx/conf.d/your_site.conf
student@lab-kubernetes:~$ sudo nano /etc/nginx/conf.d/your_site.conf
student@lab-kubernetes:~$ sudo systemctl restart nginx
student@lab-kubernetes:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
student@lab-kubernetes:~$ sudo ufw allow 'Nginx Full'
Rules updated
Rules updated (v6)
student@lab-kubernetes:~$
```

```
student@lab-kubernetes:~$ nano nginx-prometheus-exporter.yaml
student@lab-kubernetes:~$ kubectl apply -f nginx-prometheus-exporter.yaml
deployment.apps/nginx-prometheus-exporter created
```

```
student@lab-kubernetes:~$ kubectl apply -f nginx-prometheus-exporter.yaml
deployment.apps/nginx-prometheus-exporter created
student@lab-kubernetes:~$ docker run -d -p 9113:9113 --name nginx-prometheus-exporter nginx/nginx-prometheus-exporter -nginx.scrape-uri=http://
/nginx:8080/metrics
Unable to find image 'nginx/nginx-prometheus-exporter:latest' locally
latest: Pulling from nginx/nginx-prometheus-exporter
514206a90ad6: Pull complete
ab3d49933a61: Pull complete
Digest: sha256:d710e0ff2505a7037dd21e47eae07025010c0de08a6247d1a704824823becfd0
Status: Downloaded newer image for nginx/nginx-prometheus-exporter:latest
618f378fc458b35690a5f7007ca89e307e9fe8a133c8a8e0e2337726e1467d14
student@lab-kubernetes:~$
```

```
student@lab-kubernetes:~$ nano nginx-prometheus-exporter-service.yaml
student@lab-kubernetes:~$ kubectl apply -f nginx-prometheus-exporter.yaml
deployment.apps/nginx-prometheus-exporter unchanged
student@lab-kubernetes:~$
student@lab-kubernetes:~$ kubectl apply -f nginx-prometheus-exporter-service.yaml
service/promexporter created
```

```
student@lab-kubernetes:~$ wget https://get.helm.sh/helm-v3.14.4-linux-amd64.tar.gz
--2024-05-08 14:17:19-- https://get.helm.sh/helm-v3.14.4-linux-amd64.tar.gz
Resolving get.helm.sh (get.helm.sh)... 152.199.21.175, 2606:2800:233:1cb7:261b:1f9c:2074:3c
Connecting to get.helm.sh (get.helm.sh)|152.199.21.175|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 16162235 (15M) [application/x-tar]
Saving to: 'helm-v3.14.4-linux-amd64.tar.gz'
```

```
helm-v3.14.4-linux-amd64.tar.gz 100%[=====] 15.41M --KB/s in 0.08s
2024-05-08 14:17:19 (182 MB/s) - 'helm-v3.14.4-linux-amd64.tar.gz' saved [16162235/16162235]
```

```
student@lab-kubernetes:~$ tar -zxvf helm-v3.14.4-linux-amd64.tar.gz
linux-amd64/
linux-amd64/README.md
linux-amd64/LICENSE
linux-amd64/helm
student@lab-kubernetes:~$ sudo mv linux-amd64/helm /usr/local/bin/helm
student@lab-kubernetes:~$ helm version
version.BuildInfo{Version:"v3.14.4", GitCommit:"81c902a123462fd4052bc5e9aa9c513c4c8fc142", GitTreeState:"clean", GoVersion:"go1.21.9"}
```

```
student@lab-kubernetes:~$ kubectl exec -it nginx -n monitoring /bin/bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] -- [COMMAND] instead.
root@nginx:/# kubectl get namespaces
bash: kubectl: command not found
root@nginx:/# ^C
root@nginx:/# exit
exit
command terminated with exit code 130
```

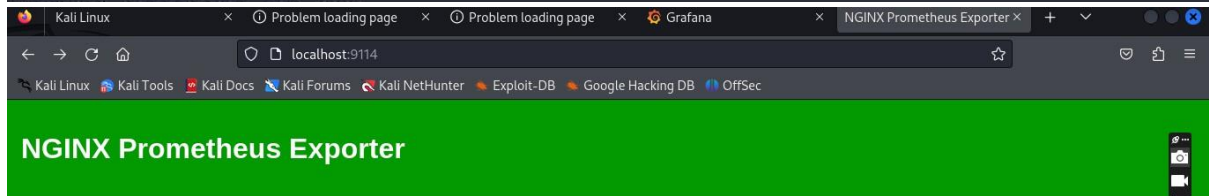
```
student@lab-kubernetes:~$ kubectl get namespaces
NAME STATUS AGE
default Active 38m
kube-node-lease Active 38m
kube-public Active 38m
kube-system Active 38m
local-path-storage Active 38m
monitoring Active 62s
```

```
student@lab-kubernetes:~$ helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
"prometheus-community" has been added to your repositories
student@lab-kubernetes:~$ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "prometheus-community" chart repository
Update Complete. ☺Happy Helming!☺
student@lab-kubernetes:~$ helm install prometheus prometheus-community/prometheus --namespace monitoring
NAME: prometheus
LAST DEPLOYED: Wed May 8 14:25:38 2024
NAMESPACE: monitoring
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
The Prometheus server can be accessed via port 80 on the following DNS name from within your cluster:
```



```
student@lab-kubernetes:~$ kubectl port-forward svc/nginx 30088:8080 -n monitoring
Forwarding from 127.0.0.1:30088 → 8080
Forwarding from [::1]:30088 → 8080
channel 13: open failed: connect failed: Connection refused
Handling connection for 30088
```

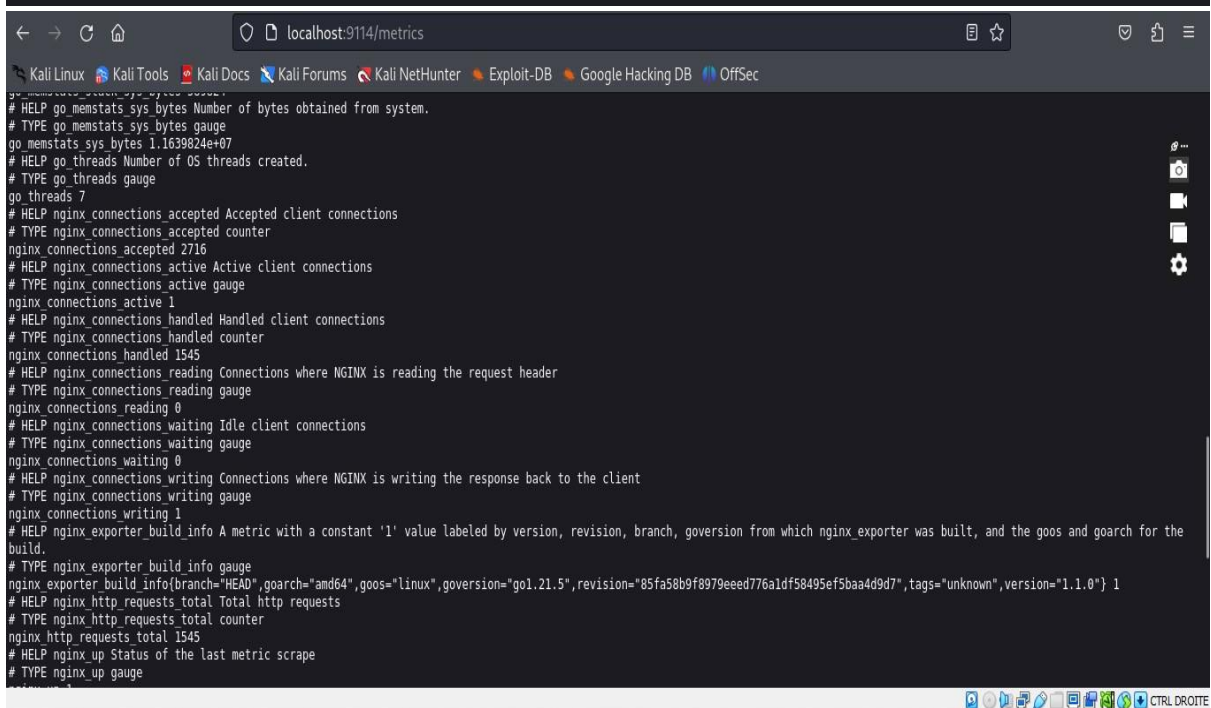
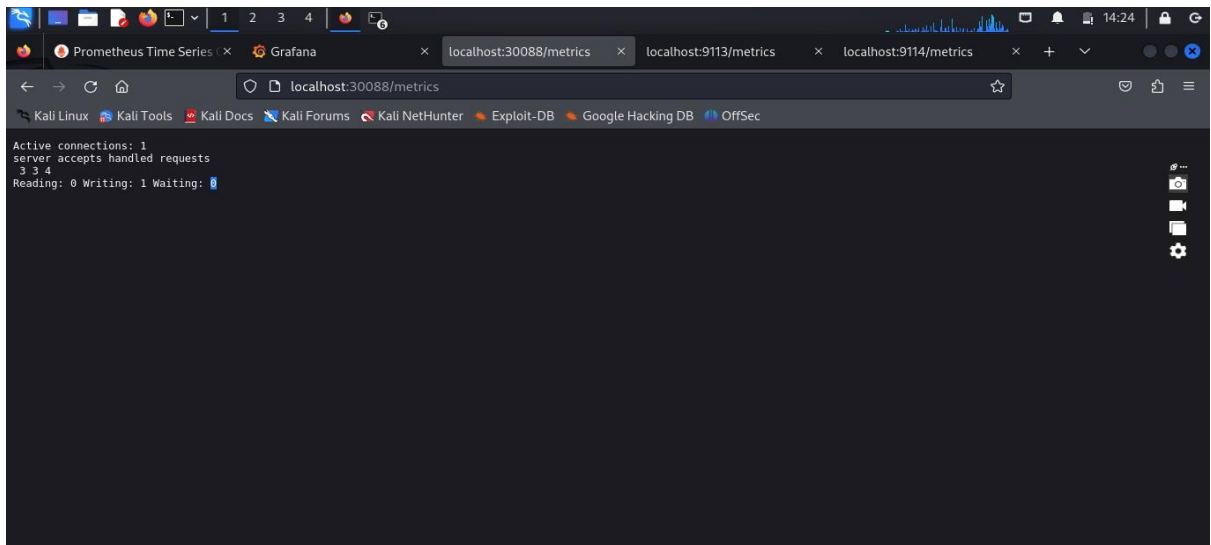
```
student@lab-kubernetes:~$ kubectl port-forward svc/promexporter 9114:9114 -n monitoring
Forwarding from 127.0.0.1:9114 → 9114
Forwarding from [::1]:9114 → 9114
channel 13: open failed: connect failed: Connection refused
Handling connection for 9114
```



Prometheus Exporter for NGINX and NGINX Plus

Version: (version=1.1.0, branch=HEAD, revision=85fa58b9f8979eed776a1dff58495ef5baa4d9d7)

- [Metrics](#)



```
Last login: Fri May 24 18:54:42 2024 from 192.168.100.254
student@lab-kubernetes:~$ kubectl port-forward svc/prometheus-server 9090:80 -n monitoring
Forwarding from 127.0.0.1:9090 -> 9090
Forwarding from [::1]:9090 -> 9090
```

```
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file will be
# reopened with the relevant failures.
```

```
#
apiVersion: v1
kind: ConfigMap
data:
  alerting_rules.yml: |
    {}
  alerts: |
    {}
  allow-snippet-annotations: "false"
  prometheus.yml: |
    global:
      scrape_interval: 1m
      evaluation_interval: 1m
      scrape_interval: 1m
      scrape_timeout: 10s
      rule_files:
        - /etc/config/recording_rules.yml
        - /etc/config/alerting_rules.yml
        - /etc/config/rules
        - /etc/config/alerts
      scrape_configs:
        - job_name: prometheus
          static_configs:
            - targets:
                - localhost:9090
          bearer_token_file: /var/run/secrets/kubernetes.io/serviceaccount/token
          job_name: kubernetes-apiservers
          kubernetes_sd_configs:
            - role: endpoints
          relabel_configs:
            - action: keep
              regex: default;kubernetes;https
```

16,1 Top

```
  - job_name: nginx
    static_configs:
      - targets:
          - localhost:9090
    bearer_token_file: /var/run/secrets/kubernetes.io/serviceaccount/token
    job_name: kubernetes-apiservers
    kubernetes_sd_configs:
      - role: endpoints
    relabel_configs:
      - action: keep
        regex: default;kubernetes;https
    source_labels:
      - __meta_kubernetes_namespace
      - __meta_kubernetes_service_name
      - __meta_kubernetes_endpoint_port_name
    scheme: https
    tls_config:
      ca_file: /var/run/secrets/kubernetes.io/serviceaccount/ca.crt
      insecure_skip_verify: true
      job_name: nginx
      static_configs:
        - targets:
            - localhost:8080
      prometheus_sd_configs:
        - job_name: prometheus
          static_configs:
            - targets:
                - localhost:9090
```

Prometheus Time Series

localhost:9090/graph?g0.expr=nginx&g0.tab=0&g0.display_mode=lines&g0.show_exemplars=0&g0.range_input=

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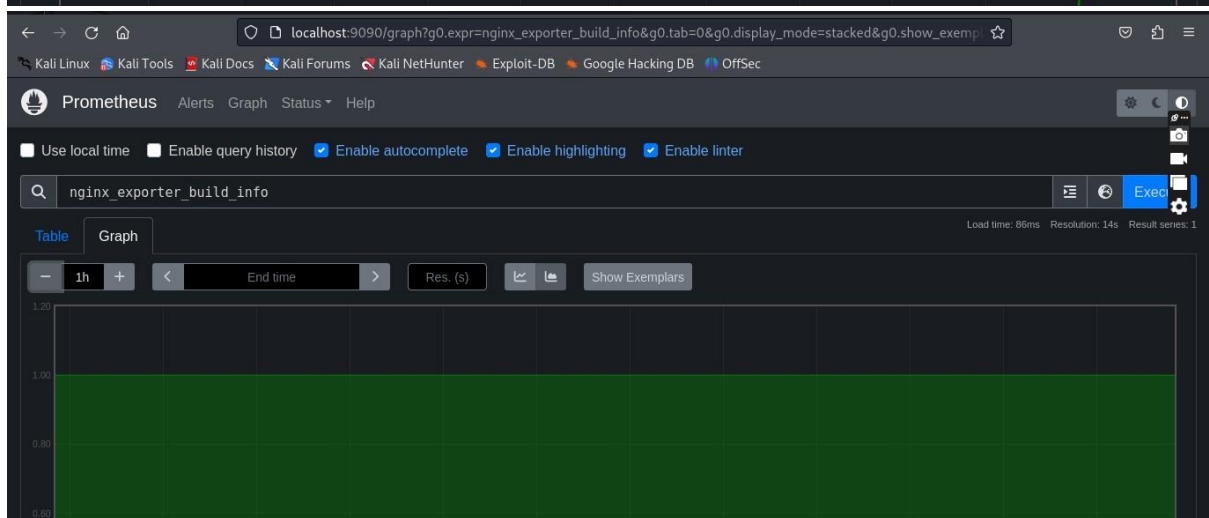
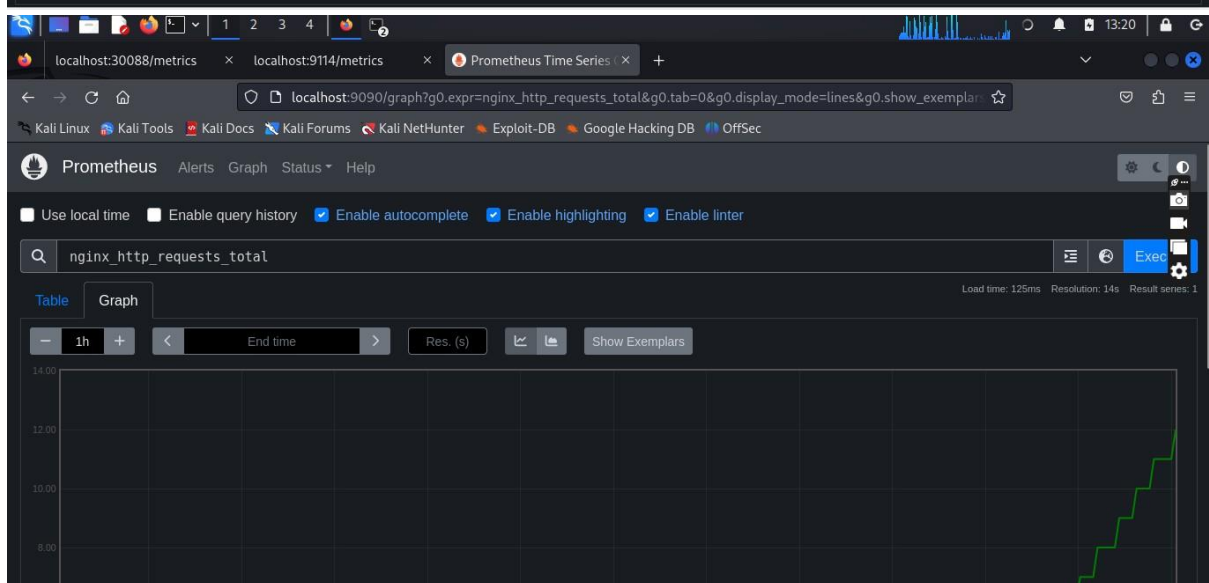
Prometheus Alerts Graph Status Help

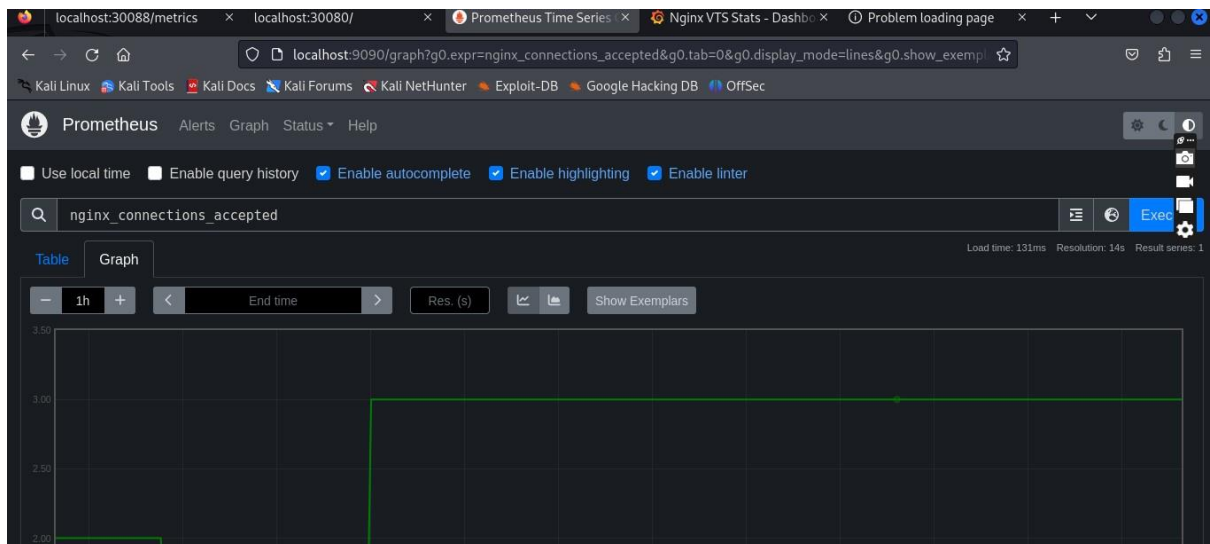
☐ Use local time ☐ Enable query history ☒ Enable autocomplete ☒ Enable highlighting ☒ Enable linter

ngi

nginx_connections_accepted	counter	Accepted client connections
nginx_connections_active	gauge	
nginx_connections_handled	counter	
nginx_connections_reading	gauge	
nginx_connections_waiting	gauge	
nginx_connections_writing	gauge	
nginx_exporter_build_info	gauge	
nginx_http_requests_total	counter	
nginx_up	gauge	
prometheus_engine_queries	gauge	
prometheus_engine_query_log_enabled	gauge	
prometheus_engine_query_samples_total	counter	
prometheus_engine_queries_concurrent_max	gauge	
prometheus_engine_query_duration_seconds	summary	
prometheus_engine_query_log_failures_total	counter	
prometheus_engine_query_duration_seconds_sum	counter	

Remove Panel





Targets

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
prometheus-agent-001	UP	instance="prometheus-agent-001"	29.418s ago	8.824ms	
prometheus-agent-002	UP	instance="prometheus-agent-002"	29.418s ago	8.824ms	
prometheus-agent-003	UP	instance="prometheus-agent-003"	29.418s ago	8.824ms	
prometheus-agent-004	UP	instance="prometheus-agent-004"	29.418s ago	8.824ms	
prometheus-agent-005	UP	instance="prometheus-agent-005"	29.418s ago	8.824ms	
prometheus-agent-006	UP	instance="prometheus-agent-006"	29.418s ago	8.824ms	
prometheus-agent-007	UP	instance="prometheus-agent-007"	29.418s ago	8.824ms	
prometheus-agent-008	UP	instance="prometheus-agent-008"	29.418s ago	8.824ms	
prometheus-agent-009	UP	instance="prometheus-agent-009"	29.418s ago	8.824ms	
prometheus-agent-010	UP	instance="prometheus-agent-010"	29.418s ago	8.824ms	
prometheus-agent-011	UP	instance="prometheus-agent-011"	29.418s ago	8.824ms	
prometheus-agent-012	UP	instance="prometheus-agent-012"	29.418s ago	8.824ms	
prometheus-agent-013	UP	instance="prometheus-agent-013"	29.418s ago	8.824ms	
prometheus-agent-014	UP	instance="prometheus-agent-014"	29.418s ago	8.824ms	
prometheus-agent-015	UP	instance="prometheus-agent-015"	29.418s ago	8.824ms	
prometheus-agent-016	UP	instance="prometheus-agent-016"	29.418s ago	8.824ms	
prometheus-agent-017	UP	instance="prometheus-agent-017"	29.418s ago	8.824ms	
prometheus-agent-018	UP	instance="prometheus-agent-018"	29.418s ago	8.824ms	
prometheus-agent-019	UP	instance="prometheus-agent-019"	29.418s ago	8.824ms	
prometheus-agent-020	UP	instance="prometheus-agent-020"	29.418s ago	8.824ms	

nginx (1/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://promexporter.default.svc.cluster.local:9114/metrics	UP	instance="promexporter.default.svc.cluster.local:9114" job="nginx"	29.418s ago	8.824ms	

prometheus (1/1 up) show more

prometheus-pushgateway (1/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://prometheus-prometheus-pushgateway.monitoring.svc:9091/metrics	UP	instance="prometheus-prometheus-pushgateway.monitoring.svc:9091" job="prometheus-pushgateway"	23.705s ago	2.319ms	

Targets

All scrape pools
All Unhealthy Collapse All
Unknown
Unhealthy
File
Settings

kubernetes-apiversers (1/1 up) [show less](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
https://172.18.0.2:6443/metrics	UP	instance="172.18.0.2:6443" job="kubernetes-apiversers"	56.95s ago	118.961ms	

kubernetes-nodes (1/1 up) [show less](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
https://kubernetes.default.svc/api/v1/nodes/kind-control-plane/proxy/metrics	UP	beta_kubernetes_io_arch="amd64" beta_kubernetes_io_os="linux"	57.776s ago	53.243ms	

kubernetes-nodes-cadvisor (1/1 up) [show less](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
https://kubernetes.default.svc/api/v1/nodes/kind-control-plane/proxy/metrics/cadvisor	UP	beta_kubernetes_io_arch="amd64" beta_kubernetes_io_os="linux" instance="kind-control-plane" job="kubernetes-nodes-cadvisor" kubernetes_io_arch="amd64" kubernetes_io_hostname="kind-control-plane" kubernetes_io_os="linux"	55.154s ago	66.794ms	

kubernetes-service-endpoints (4/4 up) [show less](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://10.244.0.10:8080/metrics	UP	app_kubernetes_io_component="metrics"	38.752s ago	3.489ms	

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localhost:9090/targets?search=promexporter

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Prometheus

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Status
Help

promexporter (1/1 up) [show less](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://promexporter.default.svc.cluster.local:9113/metrics	UP	instance="promexporter.default.svc.cluster.local:9113" job="promexporter" Discovered labels: address_="promexporter.default.svc.cluster.local:9113" metrics_path_="/metrics" scheme_="http" scrape_interval_="1m" scrape_timeout_="10s" job="promexporter"	42.115s ago	3.688ms	


```
student@lab-kubernetes:~$ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "grafana" chart repository
...Successfully got an update from the "prometheus-community" chart repository
Update Complete. 🎉Happy Helming!🎉
student@lab-kubernetes:~$ helm install grafana grafana/grafana --namespace monitoring
NAME: grafana
LAST DEPLOYED: Tue May 21 15:48:03 2024
NAMESPACE: monitoring
STATUS: deployed
REVISION: 1
NOTES:
1. Get your 'admin' user password by running:

    kubectl get secret --namespace monitoring grafana -o jsonpath="{.data.admin-password}" | base64 --decode ; echo

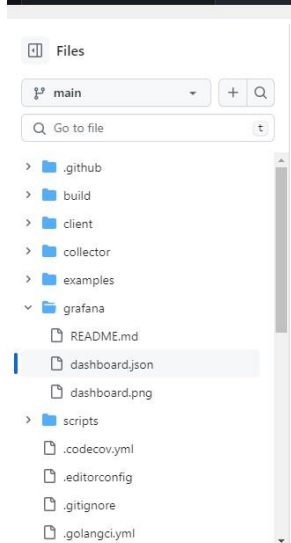
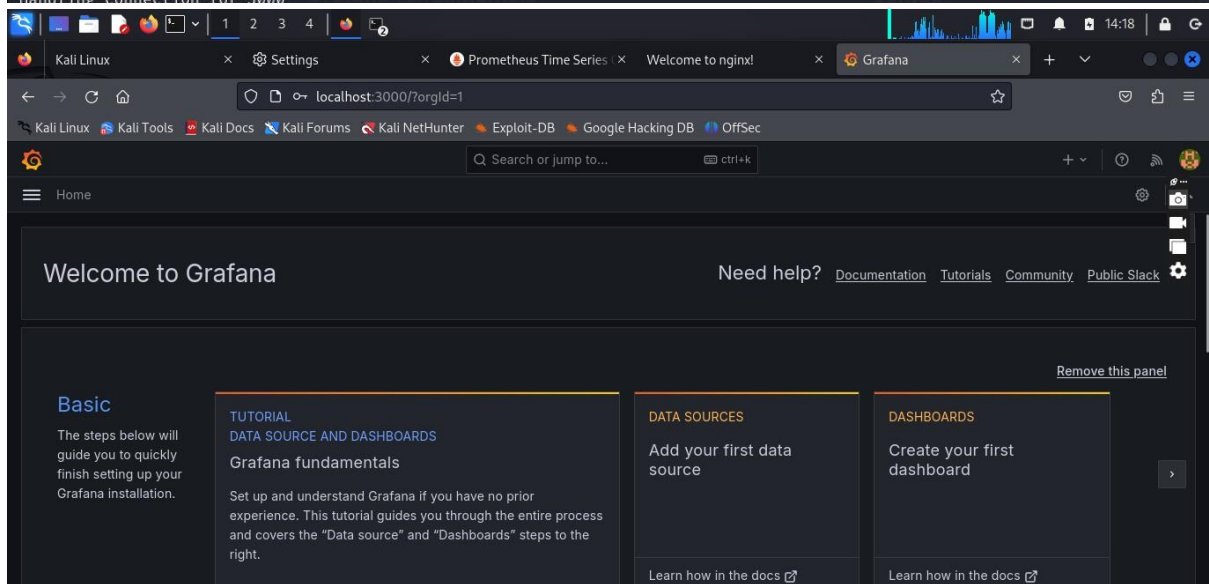
2. The Grafana server can be accessed via port 80 on the following DNS name from within your cluster:

    grafana.monitoring.svc.cluster.local

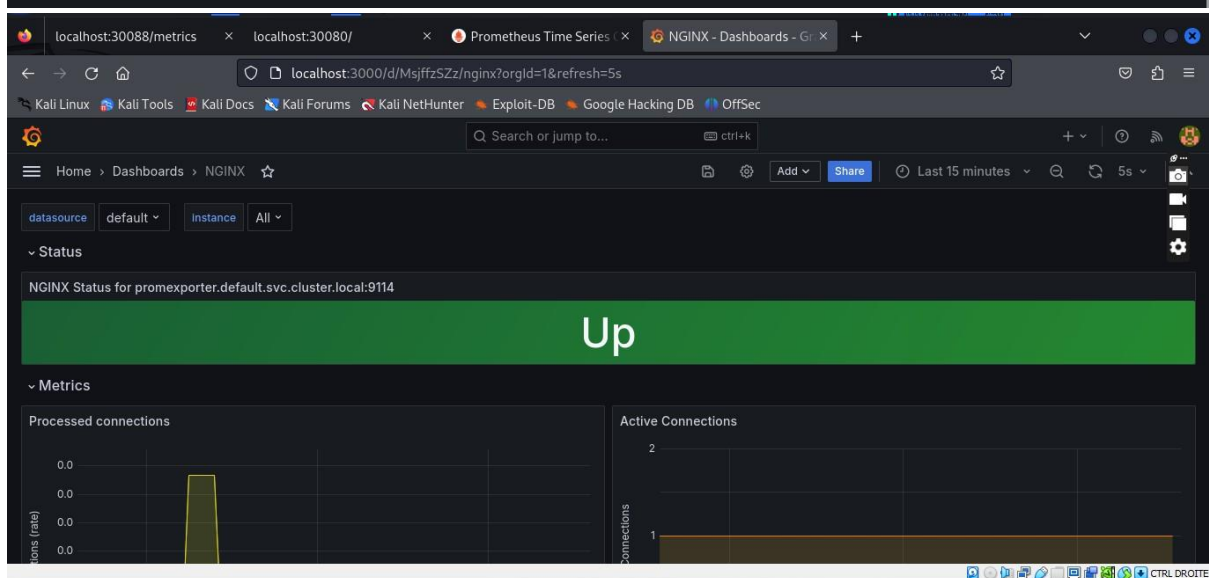
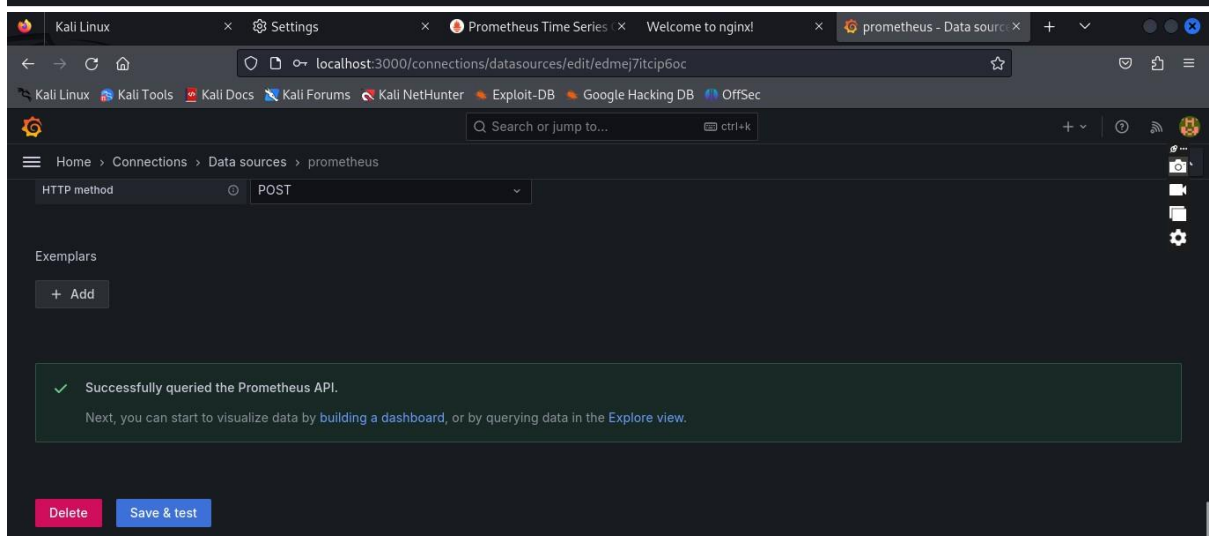
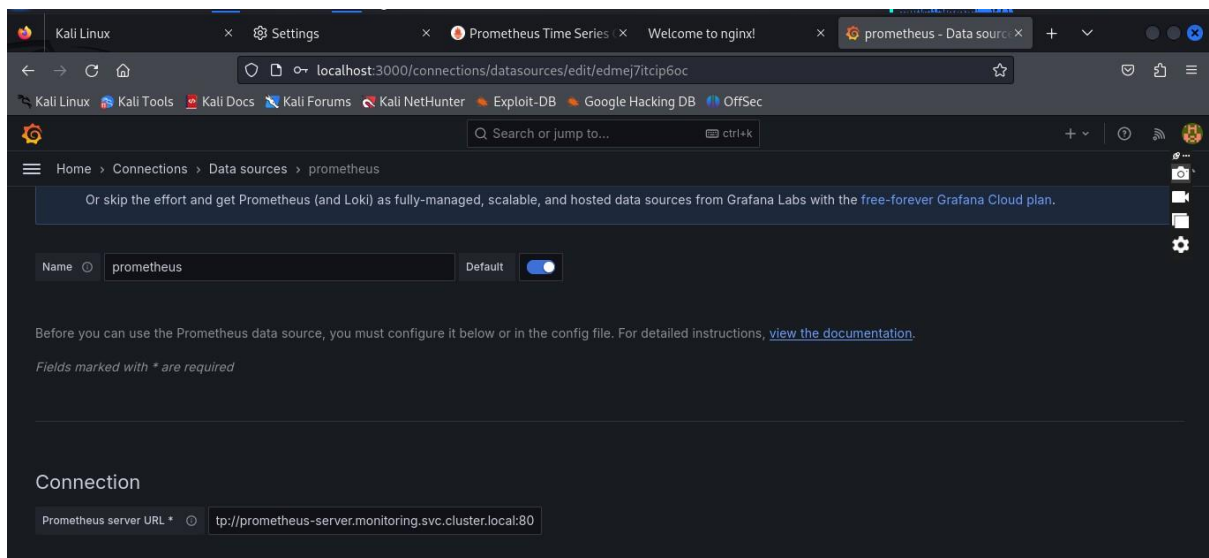
    Get the Grafana URL to visit by running these commands in the same shell:
    export POD_NAME=$(kubectl get pods --namespace monitoring -l "app.kubernetes.io/name=grafana,app.kubernetes.io/instance=grafana" -o jsonpath="{.items[0].metadata.name}")
    kubectl --namespace monitoring port-forward $POD_NAME 3000

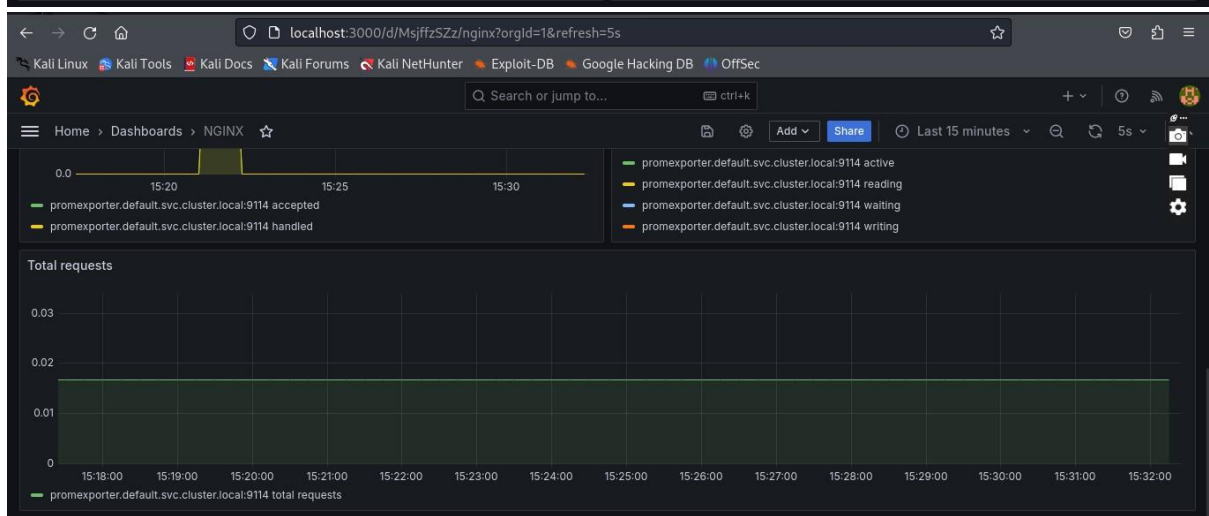
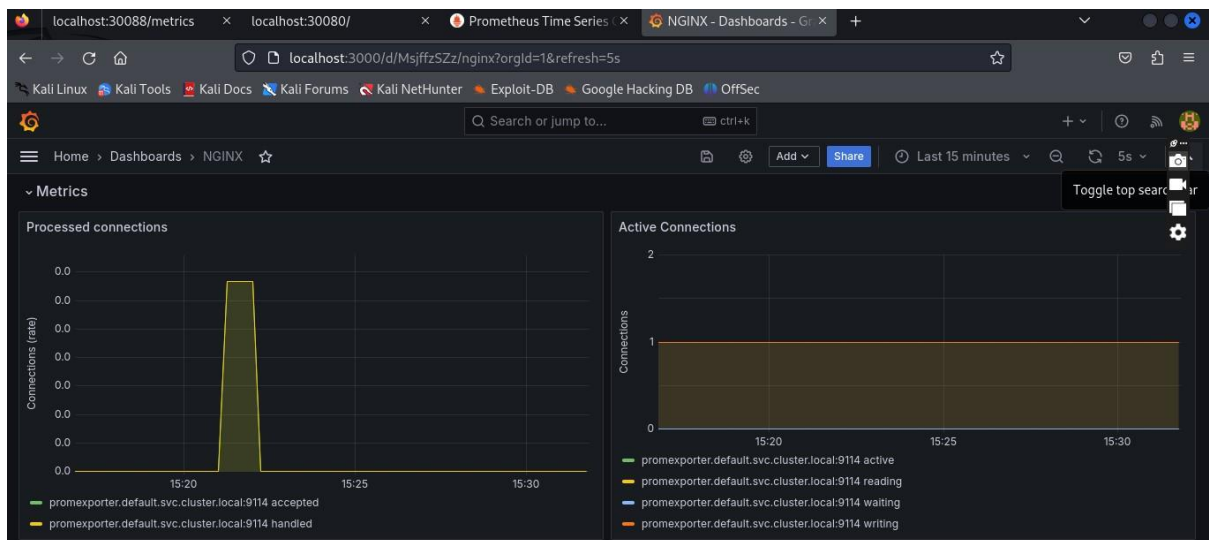
3. Login with the password from step 1 and the username: admin

#####
##### WARNING: Persistence is disabled!!! You will lose your data when #####
##### the Grafana pod is terminated. #####
#####
student@lab-kubernetes:~$ kubectl get secret --namespace monitoring grafana -o jsonpath="{.data.admin-password}" | base64 --decode ; echo
PASSWORD=$(kubectl get secret grafana-admin --namespace monitoring -o jsonpath="{.data.GF_SECURITY_ADMIN_PASSWORD}" | base64 --decode)
Password: AiRukqX0Su
student@lab-kubernetes:~$ kubectl port-forward svc/grafana 3000:3000 -n monitoring
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
```



```
nginx-prometheus-exporter / grafana / dashboard.json
lucacome Add pre-commit and fix issues (#354) 0c29bc8 · last year History
Code Blame 567 lines (567 loc) · 12.1 KB ·
1 {
2   "__inputs": [
3     {
4       "description": "",
5       "label": "Prometheus",
6       "name": "DS_PROMETHEUS",
7       "pluginId": "prometheus",
8       "pluginName": "Prometheus",
9       "type": "datasource"
10    }
11  ],
12  "__requires": [
13    {
14      "id": "grafana",
15      "name": "Grafana",
16      "type": "grafana",
17      "version": "5.0.0"
18    },
19    {
20      "id": "graph",
21      "name": "Graph",
```





```
student@lab-kubernetes: $ slowhttptest -c 1000 -H -i 10 -r 200 -t GET -u http://127.0.0.1:30080 -x 24 -p 3
```

```
File Actions Edit View Help
Thu May 23 19:52:27 2024:
slowhttptest version 1.6
- https://code.google.com/p/slowhttptest/ -
test type: SLOW HEADERS
number of connections: 1000
URL: http://127.0.0.1:30080/
Verb: GET
Content-Length header value: 4096
follow up data max size: 52
interval between follow up data: 10 seconds
connections per seconds: 200
probe connection timeout: 3 seconds
test duration: 240 seconds
using proxy: no proxy

Thu May 23 19:52:27 2024:
slow HTTP test status on 5th second:

initializing: 0
pending: 1
connected: 822
error: 0
closed: 128
service available: YES
Thu May 23 19:52:27 2024:
Test ended on 9th second
Exit status: No open connections left
student@lab-kubernetes:~$
```

nginx_up(instance="promexporter.default.svc.cluster.local:9114",job="n...")

